

The objective of this lab is to:

1. Understand and practice classes, data hiding and abstraction concepts.
2. Practice good coding conventions e.g commenting, meaningful variable and functions names, properly indented and modular code.

Instructions!

1. This is a **graded** lab, you are strictly **NOT** allowed to discuss your solutions with your fellow colleagues, even not allowed asking how is he/she is doing, it may result in negative marking. You can **ONLY** discuss with your TAs or with your instructor.
2. Strictly follow good coding conventions (commenting, meaningful variable and functions names, properly indented and modular code.
3. Save your work frequently. Make a habit of pressing **CTRL+S** after every line of code you write.

Task 01:

[20 Marks]

You are required to implement a time class. Internally, it will store the time of a day in a twenty-four-hour format. Implement the following public member functions:

```
class time
{
    int hours;
    int minutes;
    int seconds;
};
```

Note: Add all necessary checks for time.

1. **void setHour(int h);** set the current hour
2. **void setMinute(int m);** set the current minutes from 0-59
3. **void setSecond(int s);** set the current seconds
4. **void setTime(int h = 0, m = 0, s = 0);** set all three variables in a single function; **int getHour(); int getMinutes(); int getSeconds();** functions to get current time.
5. **void printTwentyFourHourFormat();** print time in 24 hour format
6. **void printTwelveHourFormat();** print time in 12 hour format
7. **void incSec(int s = 1);** Increments time by given seconds, the function should increment minutes if needed and hours if needed (use a functional approach)
8. **void incMin(int m = 1);** Increment time by given minutes the function should increment hours if needed.
9. **void incHour(int h = 1);** Increment time by given hours, reset the day if needed.

Write menu driven main program to test above class.